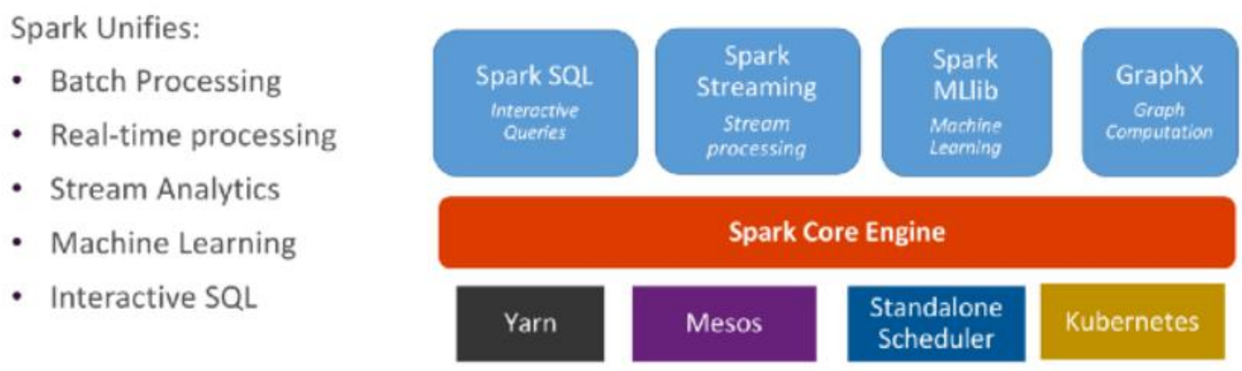
Aarya Arban

T11 – 05

Assignment No. 11




**Spark Unifies:**
Apache Spark is a unified analytics engine that supports:

- **Batch Processing** – Processing data in chunks (batches).
- **Real-time Processing** – Handling streaming data in real-time.
- **Stream Analytics** – Analyzing continuous streams of data.
- **Machine Learning** – Training and applying ML models at scale.
- **Interactive SQL** – Running SQL queries interactively on large datasets.

**Spark Core Engine (Red Block)**
At the heart of Spark is the **Spark Core Engine**, which provides:

- Basic I/O functionalities
- Memory management
- Fault recovery
- Job scheduling
- Interactions with storage systems (HDFS, S3, etc.)

**Components Built on Top of Core (Blue Blocks):**

1. **Spark SQL**
   - Allows execution of SQL queries.
   - Supports reading data from various formats like Parquet, JSON, etc.
2. **Spark Streaming**
   - For real-time stream processing (e.g., log data, sensor data).
   - Works with micro-batch architecture.
3. **Spark MLlib**
   - Machine Learning library with algorithms for classification, regression, clustering, etc.
4. **GraphX**
   - For graph processing (e.g., social network analysis, PageRank).

**Cluster Managers (Bottom Blocks):**
Spark can run on multiple cluster managers:

- **Yarn** – Hadoop’s native cluster manager.
- **Mesos** – A general-purpose cluster manager.
- **Standalone Scheduler** – Spark’s own basic cluster manager.
- **Kubernetes** – Container orchestration platform, supports Spark jobs in containers.

```
from pyspark.sql import SparkSession
```


```
from pyspark.sql import SparkSession
```


```
spark = SparkSession \
    .builder \
    .appName("Python Spark SQL basic example") \
    .config("spark.some.config.option", "some-value") \
    .getOrCreate()
```

```
spark
```

```
<div>
    <p><b>SparkSession - in-memory</b></p>
```

SparkContext

```
<p><a href="http://192.168.100.11:4040">Spark UI</a></p>

<dl>
  <dt>Version</dt>
    <dd><code>v3.5.5</code></dd>
  <dt>Master</dt>
    <dd><code>local[*]</code></dd>
  <dt>AppName</dt>
    <dd><code>Python Spark SQL basic example</code></dd>
</dl>
```

```
</div>
```

```
import os os.getcwd()
```

```
'C:\\Users\\15L\\Downloads'
```

```
df = spark.read.csv("diabetes.csv", header=True,inferSchema=True)
```

```
df.show(20)
```

```
df.head(5)
```

```
[Row(Pregnancies=6, Glucose='148', BloodPressure='72', SkinThickness=35, Insulin=0, BMI='33.6', DiabetesPedigreeFunction=0.627, Age=
 Row(Pregnancies=1, Glucose='85', BloodPressure='66', SkinThickness=29, Insulin=0, BMI='26.6', DiabetesPedigreeFunction=0.351, Age=3
  Row(Pregnancies=8, Glucose='183', BloodPressure='64', SkinThickness=0, Insulin=0, BMI='23.3', DiabetesPedigreeFunction=0.672, Age=3
  Row(Pregnancies=1, Glucose='89', BloodPressure=None, SkinThickness=23, Insulin=94, BMI=None, DiabetesPedigreeFunction=None, Age=nan
  Row(Pregnancies=0, Glucose='137', BloodPressure='40', SkinThickness=35, Insulin=168, BMI='43.1', DiabetesPedigreeFunction=2.288, Ag
```

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```
type(df)
```

```
pyspark.sql.dataframe.DataFrame
```

```
df.printSchema()
```

```
root
|-- Pregnancies: integer (nullable = true)
|-- Glucose: string (nullable = true)
|-- BloodPressure: string (nullable = true)
|-- SkinThickness: integer (nullable = true)
|-- Insulin: integer (nullable = true)
|-- BMI: string (nullable = true)
|-- DiabetesPedigreeFunction: double (nullable = true)
|-- Age: double (nullable = true)
|-- Outcome: integer (nullable = true)
```

```
df.select("Glucose").show(5)
```

```
+------------+
|Glucose|
+------------+
|    148|
|     85|
|  183|  |
89|
|    137|
+------------+
only showing top 5 rows
```

```
+-------+----------------+
df1 = df.select(df['Glucose'],df['Age']+1)
|Glucose|(Age + 1)|
+-------+----------------+
```

```
df1.show()
```

```
|      148|        51.0|
|       85|        32.0|
|      183|        33.0|
|       89|         NaN|
|      137|        34.0|
|  NULL|        31.0|
|       78|        27.0|
|          115|
             30.0| |
             197|
             54.0|
|      125|        55.0|
|      110|        31.0|
|          168|
             35.0| |
             139|
             58.0|
|      189|        60.0|
+|------- 166+|  ----------------52.0 +|
only showing top 20 |          100|
     33.0| rows
|      118|        32.0|
|       ##|        32.0|
|      103|        34.0|
|      115|        33.0|
```

```
df['Glucose']
```

```
Column<'Glucose'>
```

```
df[['Glucose','Age']]
```

```
DataFrame[Glucose: string, Age: double]
```

```
df.dtypes
```

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```
[('Pregnancies', 'int'),
 ('Glucose', 'string'),
 ('BloodPressure', 'string'),
 ('SkinThickness', 'int'),
 ('Insulin', 'int'),
 ('BMI', 'string'),
 ('DiabetesPedigreeFunction', 'double'),
 ('Age', 'double'),
 ('Outcome', 'int')]
```

```
dist_bp = df.select('BloodPressure').distinct()
```

```
dist_bp.show()
```

```
+-------------+
|BloodPressure|
+-------------+
|           54|
|           64|
|           30|
|           85|
|           52|
|            0|
|           98|
|          110|
|           96|
|          100|
|           70|
|           61|
|           75|
|           46|
|           78|
|           60|
|           90|
|           68|
|          104|
|          102|
+-------------+
only showing top 20 rows
```

```
df.select('Pregnancies').distinct().count()
```

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```
df.describe().show()
```

```
+-------+------------------+------------------+------------------+------------------+------------------+------------------+---------
|summary|       Pregnancies|      Glucose|        BloodPressure|        SkinThickness|                 Insulin|   BMI|DiabetesP
+-------+------------------+------------------+------------------+------------------+------------------+------------------+---------
|  count|               768|               767|               767|               768|               768|               767|
|   mean|3.8450520833333335|120.91906005221932| 69.07963446475196|20.536458333333332| 79.79947916666667|31.993211488250633|      0.47
| stddev| 3.36957806269887|32.009944976772395|19.363065113384383|15.952217567727642|115.24400235133803| 7.892243623420615|      0.33
|    min|                 0|                ##|                 0|                 0|                 0|                 0|
|    max|                17|                99|                 ?|                99|               846|                 ?|
+-------+------------------+------------------+------------------+------------------+------------------+------------------+---------
```

```
df.describe(['BMI','Glucose']).show()
```

```
+-------+------------------+------------------+
|summary|               BMI|           Glucose|
+-------+------------------+------------------+
| count|               767|               767|
| mean|31.993211488250633|120.91906005221932|
| stddev| 7.892243623420615|32.009944976772395|
|    min|                 0|                ##|
```

```
|    max|                 ?|                99|
+-------+------------------+ --------------------------------- +
```

```
df.describe(['Age']).show()
```

```
+-------+ ------ +
|summary| Age|
+-------+ ------ +
| count| 768|
| mean| NaN|
| stddev| NaN|
|    min|21.0|
|    max| NaN|
+-------+ ------ +
```

```
df['Age']
```

```
Column<'Age'>
```

```
#adding column
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------+ ------------- +
df_new = df.withColumn('New Age',df.Age+2)
|Pregnancies|Glucose|BloodPressure|SkinThickness|Insulin| BMI|DiabetesPedigreeFunction| Age|Outcome|New Age|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------+ ------------- +

df_new.show()
```

```
|           6|      148|              72|              35|          0|33.6|                         0.627|50.0|        1|      52.0|
|           1|       85|              66|              29|          0|26.6|                         0.351|31.0|        0|      33.0|
|           8|      183|              64|               0|          0|23.3|                         0.672|32.0|        1|      34.0|
|           1|       89|            NULL|              23|         94|NULL|                          NULL| NaN|        0|       NaN|
|           0|      137|              40|              35|        168|43.1|                         2.288|33.0|        1|      35.0|
|           5|     NULL|              74|               0|          0|25.6|                         0.201|30.0|        0|      32.0|
|           3|       78|              50|              32|         88| 31|                          0.248|26.0|        1|      28.0|
|          10|      115|               0|               0|          0|35.3|                         0.134|29.0|        0|      31.0|
|           2|      197|              70|              45|        543|30.5|                         0.158|53.0|        1|      55.0|
|           8|      125|              96|               0|          0|     0|                       0.232|54.0|        1|      56.0|
|           4|      110|               ?|               0|          0|37.6|                         0.191|30.0|        0|      32.0|
|          10|      168|              74|               0|          0| 38|                          0.537|34.0|        1|      36.0|
|          10|      139|              80|               0|          0|27.1|                         1.441|57.0|        0|      59.0|
|           1|      189|              60|              23|        846|30.1|                         0.398|59.0|        1|      61.0|
+| -----------5+|------- 166+|-------------   72+|-------------      19+|------- 175|25.8+----+|------------------------     0.587|51.0+--
--+|-------            1+| -------------        53.0| +

only showing top 20  | 7|         100rows|           0|             0|                           0| 30|      0.484|32.0| 1|   34.0|
|           0|      118|              84|              47|        230|45.8|                         0.551|31.0|        1|      33.0|
|           7|       ##|              74|               0|          0|29.6|                         0.254|31.0|        1|      33.0|
|           1|      103|              30|              38|         83|43.3|                         0.183|33.0|        0|      35.0|
|           1|      115|              70|              30|         96|34.6|                         0.529|32.0|        1|      34.0|
```

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```
#df.select(isnull())
```

```
+-----------+-------+-------------+-------------+-------+----+------------------------+----+ ------------ +
df_new= df_new.drop('New Age')
|Pregnancies|Glucose|BloodPressure|SkinThickness|Insulin| BMI|DiabetesPedigreeFunction| Age|Outcome|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+ ------------ +

df_new.show()

|          6|    148|           72|           35|      0|33.6|                   0.627|50.0|      1|
|          1|     85|           66|           29|      0|26.6|                   0.351|31.0|      0|
|          8|    183|           64|            0|      0|23.3|                   0.672|32.0|      1|
|          1|     89|         NULL|           23|     94|NULL|                    NULL| NaN|      0|
|          0|    137|           40|           35|    168|43.1|                   2.288|33.0|      1|
|          5|   NULL|           74|            0|      0|25.6|                   0.201|30.0|      0|
|          3|     78|           50|           32|     88| 31|                    0.248|26.0|      1|
|         10|    115|            0|            0|      0|35.3|                   0.134|29.0|      0|
|          2|    197|           70|           45|    543|30.5|                   0.158|53.0|      1|
|          8|    125|           96|            0|      0|   0|                   0.232|54.0|      1|
|          4|    110|            ?|            0|      0|37.6|                   0.191|30.0|      0|
|         10|    168|           74|            0|      0| 38|                    0.537|34.0|      1|
|         10|    139|           80|            0|      0|27.1|                   1.441|57.0|      0|
|          1|    189|           60|           23|    846|30.1|                   0.398|59.0|      1|
+|-----------5+|------- 166+|-------------   72+|-------------    19+|------- 175|25.8+----+|------------------------    0.587|51.0+---
-+|------------                1 +|

only showing top 20 |7|    100rows|            0|            0|      0| 30|   0.484|32.0|      1|
|          0|    118|           84|           47|    230|45.8|                   0.551|31.0|      1|
|          7|     ##|           74|            0|      0|29.6|                   0.254|31.0|      1|
|          1|    103|           30|           38|     83|43.3|                   0.183|33.0|      0|
|          1|    115|           70|           30|     96|34.6|                   0.529|32.0|      1|
```

```
df_null = df.filter(df['BloodPressure'].isNull())
```

```
df_null.show()
```

```
+-----------+-------+-------------+-------------+-------+----+------------------------+---+ ------------ +
|Pregnancies|Glucose|BloodPressure|SkinThickness|Insulin| BMI|DiabetesPedigreeFunction|Age|Outcome|
+-----------+-------+-------------+-------------+-------+----+------------------------+---+ ------------ +
|          1|     89|         NULL|           23|     94|NULL|                    NULL|NaN|      0|
+-----------+-------+-------------+-------------+-------+----+------------------------+---+ ------------ +
```

```
df.select('Pregnancies').distinct().show()
```

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|Pregnancies|

# ssignment +--------------------+ 11: EDA using Apache spark & Pandas

```
|          12|
|           1|
|          13|
|           6|
|        3| |
        5|
|          15|
|           9|
|          17|
|           4|
|           8|
+| --------------------7+|
```

T

```
|      10| |
      11|
|          14|
|           2|
|           0|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+ -------------
+
|Pregnancies|Glucose|BloodPressure|SkinThickness|Insulin| BMI|DiabetesPedigreeFunction| Age|Outcome|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+ -------------
+
```

```
df.show(20)
```

```
|          6|    148|           72|           35|      0|33.6|                   0.627|50.0|       1|
|          1|     85|           66|           29|      0|26.6|                   0.351|31.0|       0|
|          8|    183|           64|            0|      0|23.3|                   0.672|32.0|       1|
|          1|     89|         NULL|           23|     94|NULL|                    NULL| NaN|       0|
|          0|    137|           40|           35|    168|43.1|                   2.288|33.0|       1|
|          5|   NULL|           74|            0|      0|25.6|                   0.201|30.0|       0|
| 3| 78| 50| 32| 88| 31| 0.248|26.0| 1| | 10| 115| 0| 0| 0|35.3| 0.134|29.0| 0|
|          2|    197|           70|           45|    543|30.5|                   0.158|53.0|       1|
|          8|    125|           96|            0|      0|   0|                   0.232|54.0|       1|
|          4|    110|            ?|            0|      0|37.6|                   0.191|30.0|       0|
|         10|    168|           74|            0|      0|  38|                   0.537|34.0|       1|
|         10|    139|           80|            0|      0|27.1|                   1.441|57.0|       0|
|          1|    189|           60|           23|    846|30.1|                   0.398|59.0|       1|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------------
+
|      5|      166|     72|      19|      175|25.8|   0.587|51.0| 1| only showing top 20 rows
|          7|    100|            0|            0|      0|  30|                   0.484|32.0|       1|
|          0|    118|           84|           47|    230|45.8|                   0.551|31.0|       1|
|          7|     ##|           74|            0|      0|29.6|                   0.254|31.0|       1|
|          1|    103|           30|           38|     83|43.3|                   0.183|33.0|       0|
|          1|    115|           70|           30|     96|34.6|                   0.529|32.0|       1|
```

```
df1 = df.na.drop(how='any')
```

```
df1.show(20)
```

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```
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------------+
|Pregnancies|Glucose|BloodPressure|SkinThickness|Insulin| BMI|DiabetesPedigreeFunction| Age|Outcome|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------------+
```

# Assignment 11: EDA using Apache spark & Pandas

```
|          6|    148|           72|           35|      0|33.6|                   0.627|50.0|       1|
|          1|     85|           66|           29|      0|26.6|                   0.351|31.0|       0|
|          8|    183|           64|            0|      0|23.3|                   0.672|32.0|       1|
|          0|    137|           40|           35|    168|43.1|                   2.288|33.0|       1|
|          3|     78|           50|           32|     88|  31|                   0.248|26.0|       1|
|         10|    115|            0|            0|      0|35.3|                   0.134|29.0|       0|
|          2|    197|           70|           45|    543|30.5|                   0.158|53.0|       1|
|          8|    125|           96|            0|      0|   0|                   0.232|54.0|       1|
|          4|    110|            ?|            0|      0|37.6|                   0.191|30.0|       0|
|         10|    168|           74|            0|      0|  38|                   0.537|34.0|       1|
|         10|    139|           80|            0|      0|27.1|                   1.441|57.0|       0|
|          1|    189|           60|           23|    846|30.1|                   0.398|59.0|       1|
|          5|    166|           72|           19|    175|25.8|                   0.587|51.0|       1|
|          7|    100|            0|            0|      0|  30|                   0.484|32.0|       1|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------------+
|          0|    118|           84|           47|    230|45.8|                   0.551|31.0|       1|
only showing top 20 rows
|          7|     ##|           74|            0|      0|29.6|                   0.254|31.0|       1|
|          1|    103|           30|           38|     83|43.3|                   0.183|33.0|       0|
|          1|    115|           70|           30|     96|34.6|                   0.529|32.0|       1|
|          3|    126|           88|           41|    235|39.3|                   0.704|27.0|       0|
|          8|     99|           84|            0|      0|   ?|                   0.388|50.0|       0|
```

```
+-----------+-------+-------------+-------------+-------+----+------------------------+----+------------- +
df1 = |Pregnancies|Glucose|BloodPressure|SkinThickness|Insulin| BMI|DiabetesPedigreeFunction| df.na.drop(how='any',thresh=7)
Age|Outcome|
-------------
+-----------+-------+-------------+-------------+-------+----+------------------------+----+          +
```

```
df1.show(20)
```

```
|          6|    148|           72|           35|      0|33.6|                   0.627|50.0|      1|
|          1|     85|           66|           29|      0|26.6|    0.351|31.0|  0| |          8|    183|           64|            0|      0|23.3|
0.672|32.0|  1|
|          0|    137|           40|           35|    168|43.1|                   2.288|33.0|      1|
|          5|   NULL|           74|            0|      0|25.6|                   0.201|30.0|      0|
| 3| 78| 50| 32| 88| 31| 0.248|26.0| 1| | 10| 115| 0| 0| 0|35.3| 0.134|29.0| 0|
|          2|    197|           70|           45|    543|30.5|                   0.158|53.0|      1|
|          8|    125|           96|            0|      0|   0|                   0.232|54.0|      1|
|          4|    110|            ?|            0|      0|37.6|                   0.191|30.0|      0|
|         10|    168|           74|            0|      0|  38|                   0.537|34.0|      1|
|         10|    139|           80|            0|      0|27.1|                   1.441|57.0|      0|
|          1|    189|           60|           23|    846|30.1|                   0.398|59.0|      1|
|          5|    166|           72|           19|    175|25.8|                   0.587|51.0|      1|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------+
|          7|    100|            0|            0|      0|  30|    0.484|32.0|  1| only showing top 20 rows
|          0|    118|           84|           47|    230|45.8|                   0.551|31.0|      1|
|          7|     ##|           74|            0|      0|29.6|                   0.254|31.0|      1|
|          1|    103|           30|           38|     83|43.3|                   0.183|33.0|      0|
|          1|    115|           70|           30|     96|34.6|                   0.529|32.0|      1|
|          3|    126|           88|           41|    235|39.3|                   0.704|27.0|      0|
```

```
df1 =
df.na.drop(how='any',subset=['Glucose'])
```

```
df1.show(20)
```

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# Assignment 11: EDA using Apache spark & Pandas

```
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------+
|Pregnancies|Glucose|BloodPressure|SkinThickness|Insulin| BMI|DiabetesPedigreeFunction| Age|Outcome|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------+
|          6|    148|           72|           35|      0|33.6|                   0.627|50.0|      1|
|          1|     85|           66|           29|      0|26.6|                   0.351|31.0|      0|
|          8|    183|           64|            0|      0|23.3|                   0.672|32.0|      1|
|          1|     89|         NULL|           23|     94|NULL|                    NULL| NaN|      0|
|          0|    137|           40|           35|    168|43.1|                   2.288|33.0|      1|
|          3|     78|           50|           32|     88|  31|                   0.248|26.0|      1|
|         10|    115|            0|            0|      0|35.3|                   0.134|29.0|      0|
|          2|    197|           70|           45|    543|30.5|                   0.158|53.0|      1|
|          8|    125|           96|            0|      0|   0|                   0.232|54.0|      1|
|          4|    110|            ?|            0|      0|37.6|                   0.191|30.0|      0|
|         10|    168|           74|            0|      0|  38|                   0.537|34.0|      1|
|         10|    139|           80|            0|      0|27.1|                   1.441|57.0|      0|
|          1|    189|           60|           23|    846|30.1|                   0.398|59.0|      1|
|          5|    166|           72|           19|    175|25.8|                   0.587|51.0|      1|
|          7|    100|            0|            0|      0|  30|                   0.484|32.0|      1|
|          0|    118|           84|           47|    230|45.8|                   0.551|31.0|      1|
|          7|     ##|           74|            0|      0|29.6|                   0.254|31.0|      1|
|          1|    103|           30|           38|     83|43.3|                   0.183|33.0|      0|
|          1|    115|           70|           30|     96|34.6|                   0.529|32.0|      1|
|          3|    126|           88|           41|    235|39.3|                   0.704|27.0|      0|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------+
only showing top 20 rows
```

```
df1 = df.na.fill("Missing",subset=["BloodPressure","Glucose"])
```

```
df1.show(20)
```

```
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------------+
|Pregnancies|Glucose|BloodPressure|SkinThickness|Insulin| BMI|DiabetesPedigreeFunction| Age|Outcome|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------------+
|          6|    148|           72|           35|      0|33.6|                   0.627|50.0|      1|
|          1|     85|           66|           29|      0|26.6|                   0.351|31.0|      0|
|          8|    183|           64|            0|      0|23.3|                   0.672|32.0|      1|
|          1|     89|      Missing|           23|     94|NULL|                    NULL| NaN|      0|
|          0|    137|           40|           35|    168|43.1|                   2.288|33.0|      1|
|          5|Missing|           74|            0|      0|25.6|                   0.201|30.0|      0|
|          3|     78|           50|           32|     88|  31|                   0.248|26.0|      1|
|         10|    115|            0|            0|      0|35.3|                   0.134|29.0|      0|
|          2|    197|           70|           45|    543|30.5|                   0.158|53.0|      1|
|          8|    125|           96|            0|      0|   0|                   0.232|54.0|      1|
|          4|    110|            ?|            0|      0|37.6|                   0.191|30.0|      0|
|         10|    168|           74|            0|      0|  38|                   0.537|34.0|      1|
|         10|    139|           80|            0|      0|27.1|                   1.441|57.0|      0|
|          1|    189|           60|           23|    846|30.1|                   0.398|59.0|      1|
|          5|    166|           72|           19|    175|25.8|                   0.587|51.0|      1|
|          7|    100|            0|            0|      0|  30|                   0.484|32.0|      1|
|          0|    118|           84|           47|    230|45.8|                   0.551|31.0|      1|
|          7|     ##|           74|            0|      0|29.6|                   0.254|31.0|      1|
|          1|    103|           30|           38|     83|43.3|                   0.183|33.0|      0|
|          1|    115|           70|           30|     96|34.6|                   0.529|32.0|      1|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------------+
only showing top 20 rows
```

```
from pyspark.ml.feature import Imputer
```

```
imputer = Imputer(inputCols=['Age'], outputCols=['Imputed Age'],strategy="mean")
```

## ssignment 11: EDA using Apache spark & Pandas

```
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------+-------------------------------+
|Pregnancies|Glucose|BloodPressure|SkinThickness|Insulin| BMI|DiabetesPedigreeFunction| Age|Outcome|        Imputed Age|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------+-------------------------------+
```

```
imputer.fit(df).transform(df).show(20)
```

```
|          6|    148|           72|           35|      0|33.6|                   0.627|50.0|      1|               50.0|
|          1|     85|           66|           29|      0|26.6|                   0.351|31.0|      0|               31.0|
|          8|    183|           64|            0|      0|23.3|                   0.672|32.0|      1|               32.0|
|          1|     89|         NULL|           23|     94|NULL|                    NULL| NaN|      0|33.25684485006519|
|          0|    137|           40|           35|    168|43.1|                   2.288|33.0|      1|               33.0|
|          5|   NULL|           74|            0|      0|25.6|                   0.201|30.0|      0|               30.0|
|          3|     78|           50|           32|     88| 31|                   0.248|26.0|      1|               26.0|
|         10|    115|            0|            0|      0|35.3|                   0.134|29.0|      0|               29.0|
|          2|    197|           70|           45|    543|30.5|                   0.158|53.0|      1|               53.0|
|          8|    125|           96|            0|      0|   0|                   0.232|54.0|      1|               54.0|
|          4|    110|            ?|            0|      0|37.6|                   0.191|30.0|      0|               30.0|
|         10|    168|           74|            0|      0| 38|                   0.537|34.0|      1|               34.0|
|         10|    139|           80|            0|      0|27.1|                   1.441|57.0|      0|               57.0|
|          1|    189|           60|           23|    846|30.1|                   0.398|59.0|      1|               59.0|
+|-----------5+|------- 166+|-------------   72+|-------------    19+|------- 175|25.8+----+|-------------------------    0.587|51.0+---
-+|-------    1+|  -------------------------------51.0| +
only showing top 20  | 7|        100rows|           0|           0|                        0| 30|   0.484|32.0|          1|   32.0|
|          0|    118|           84|           47|    230|45.8|                   0.551|31.0|      1|               31.0|
|          7|     ##|           74|            0|      0|29.6|                   0.254|31.0|      1|               31.0|
|          1|    103|           30|           38|     83|43.3|                   0.183|33.0|      0|               33.0|
|          1|    115|           70|           30|     96|34.6|                   0.529|32.0|      1|               32.0|
```

```
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------------+
|Pregnancies|Glucose|BloodPressure|SkinThickness|Insulin| BMI|DiabetesPedigreeFunction| Age|Outcome|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------------+
```

```
df.filter("Age>50").show()
```

```
|          1|     89|         NULL|           23|     94|NULL|                    NULL| NaN|      0|
|          2|    197|           70|           45|    543|30.5|                   0.158|53.0|      1|
|          8|    125|           96|            0|      0|   0|                   0.232|54.0|      1|
|         10|    139|           80|            0|      0|27.1|                   1.441|57.0|      0|
|     1|       189|     60|       23|       846|30.1|   0.398|59.0| 1| |      5|       166|      72|
      19|       175|25.8|   0.587|51.0| 1|
|         11|    143|           94|           33|    146|36.6|                   0.254|51.0|      1|
|         13|    145|           82|           19|    110|22.2|                   0.245|57.0|      0|
|          5|    109|           75|           26|      0| 36|                   0.546|60.0|      0|
|          4|    111|           72|           47|    207|37.1|                    1.39|56.0|      1|
|          9|    171|          110|           24|    240|45.4|                   0.721|54.0|      1|
|          8|    176|           90|           34|    300|33.7|                   0.467|58.0|      1|
|          2|    109|           92|            0|      0|42.7|                   0.845|54.0|      0|
|          4|    134|           72|            0|      0|23.8|                   0.277|60.0|      1|
+-----------+-------+-------------+-------------+-------+----+------------------------+----+-------------+
|     4|       146|     92|        0|         0|31.2|     0.539|61.0| 1| only showing top 20 rows
|          5|    132|           80|            0|      0|26.8|                   0.186|69.0|      0|
|          0|    105|           84|            0|      0|27.9|                   0.741|62.0|      1|
|          3|    128|           78|            0|      0|21.1|                   0.268|55.0|      0|
|          5|    147|           78|            0|      0|33.7|                   0.218|65.0|      0|
|          8|    181|           68|           36|    495|30.1|                   0.615|60.0|      1|
```

```
df.filter("Age>50").select(['Age','Insulin','Glucose','Outcome']).show()
```

```
11+----- +0-------6 +--------+ ------------ +
| Age|Insulin|Glucose|Outcome|
```

ssignment +----+-------+-------11:+ EDA using Apache spark & Pandas------------ +

```
| NaN|      94|     89|       0|
|53.0|     543|    197|       1|
|54.0|       0|    125|       1|
|57.0|       0|    139|       0|
|59.0|  846|        189|       1|
|51.0|  175|        166|       1|
|51.0|     146|    143|       1|
|57.0|     110|    145|       0|
|60.0|       0|    109|       0|
|56.0|     207|    111|       1|
|54.0|     240|    171|       1|
|58.0|     300|    176|       1|
|54.0|       0|    109|       0|
|60.0|       0|    134|       1|
+|61.0|----+ -------0+|------- 146|+ ------------1 +|
only showing top 20 |69.0|     0|       132rows|   0|
```

T

```
|62.0| 0| 105| 1| |55.0| 0| 128| 0|
|65.0|       0|    147|       0|
|60.0|     495|    181|       1|
```

```
+----+-------+-------+ ------------ +
| Age|Insulin|Glucose|Outcome|
+----+-------+-------+ ------------ +
```

```
df.filter(~(df['Age']>50) & (df['Glucose']>175)).select(['Age','Insulin','Glucose','Outcome']).show()
```

```
|32.0|      0|    183|      1|
|41.0|      0|    196|      1|
|26.0|     70|    180|      0|
|25.0|      0|    180|      1|
|41.0|    304|    187|      1|
|43.0|      0|    188|      1|
|36.0|    130|    179|      1|
|41.0|      0|    194|      1|
|41.0|      0|    184|      1|
|21.0|    478|    177|      1|
|31.0|    744|    197|      0|
|23.0|      0|    179|      1|
|49.0|      0|    184|      1|
|24.0|    375|    193|      0|
|34.0|    130|    191|      0|
|29.0|      0|    182|      1|
|37.0|      0|    179|      0|
|41.0|      0|    180|      1|
|41.0|      0|    178|      1|
|29.0|    249|    196|      1|
+----+-------+-------+-------+
only showing top 20 rows
```

# ssignment 11: EDA using Apache spark & Pandas

```
df.filter((df['Age']>50) & (df['Glucose']>175)).select(['Age','Insulin','Glucose','Outcome']).show()
```

```
+----+-------+-------+-------+
| Age|Insulin|Glucose|Outcome|
+----+-------+-------+-------+
|53.0|    543|    197|      1|
|59.0|    846|    189|      1|
|58.0|    300|    176|      1|
|60.0|    495|    181|      1|
|57.0|    280|    196|      1|
|60.0|      0|    179|      0|
|51.0|    192|    181|      1|
|59.0|      0|    194|      1|
|67.0|      0|    194|      0|
|55.0|    145|    195|      1|
|53.0|    207|    187|      1|
|62.0|      0|    197|      1|
|52.0|    156|    176|      1|
|66.0|      0|    190|      1|
+----+-------+-------+-------+
```

```
df.groupBy(['Age']).count().show()
```

```
+----+-----+
| Age|count|
+----+-----+
```

```
|67.0|    3| |70.0|
          1|
|69.0|    2|
| NaN|    1|
|49.0|    5|
|29.0|   29|
|64.0|    1|
|47.0|    6|
|42.0|   18|
|44.0|    8|
|35.0|   10|
|62.0|    4|
|39.0|   12|
|34.0|   14|
+----+ --------- +
|37.0|   19|
only showing top 20 rows
|25.0|   48|
|36.0|   16|
|41.0|   22|
|23.0|   38|
|50.0|    8|
```

```
df.groupBy('BloodPressure').max().show()
```

```
+-------------+----------------+-----------------+------------+-----------------------------+--------+------------+
|BloodPressure|max(Pregnancies)|max(SkinThickness)|max(Insulin)|max(DiabetesPedigreeFunction)|max(Age)|max(Outcome)|
+-------------+----------------+-----------------+------------+-----------------------------+--------+------------+
|           54|               7|               32|         175|                        0.748|    62.0|           1|
|           64|               8|               46|         415|                        1.731|    41.0|           1|
|           30|               1|               42|          99|                        0.496|    33.0|           1|
|           85|              12|               54|         100|                        1.213|    56.0|           1|
|           52|               9|               43|         540|                        1.699|    42.0|           1|
|            0|              13|               30|           0|                        0.933|    72.0|           1|
|           98|              10|               41|          58|                        1.321|    34.0|           1|
|          110|               9|               46|         240|                        0.721|    54.0|           1|
|         NULL|               1|               23|          94|                         NULL|     NaN|           0|
|           96|               8|                0|           0|                        0.268|    54.0|           1|
|          100|               8|               39|         240|                        1.021|    45.0|           1|
|           70|              15|               99|         744|                        2.329|    63.0|           1|
|           61|               6|                0|           0|                        0.151|    55.0|           0|
|           75|               5|               32|           0|                        0.572|    60.0|           1|
+-------------+----------------+-----------------+------------+-----------------------------+--------+------------+
|           46|               2|               21|         335|                        0.654|    22.0|           0|
only showing top 20 rows
|           78|              14|               63|         293|                         2.42|    67.0|           1|
|           60|              13|               46|         846|                        1.072|    67.0|           1|
|           90|              13|               51|         680|                        0.805|    66.0|           1|
|           68|              11|               49|         579|                        1.391|    60.0|           1|
|          104|               5|               25|           0|                        0.435|    52.0|           1|
```

```
df.groupBy('BloodPressure').mean().show()
```

```
|           60|2.6486486486486487| 20.18918918918919|105.70270270270271|   0.43645945945945945|29.135135135135137| 0.18918918
|           90| 4.545454545454546|23.545454545454547|116.13636363636364|   0.4235454545454545| 38.13636363636363|
|           68| 3.488888888888889|23.133333333333333| 89.08888888888889|  0.47006666666666674|31.555555555555557|0.266666666
|          104|               2.5|              12.5|               0.0|                0.293|              46.5|
```

```
df.agg({"Insulin":'max'}).show()
```

```
+ -------------------- +
|max(Insulin)|
+ -------------------- +
|         846|
+ -------------------- +
```

| BloodPressure | avg(Pregnancies) | avg(SkinThickness) | avg(Insulin) | avg(DiabetesPedigreeFunction) | avg(Age) | avg( |
|---|---|---|---|---|---|---|
| 54 | 2.727272727272727 | 16.363636363636363 | 76.81818181818181 | 0.43290909090909097 | 29.181818181818183 | 0.181818181 |
| 64 | 2.4186046511627906 | 23.348837209302324 | 90.13953488372093 | 0.48055813953488374 | 26.13953488372093 | 0.30232558 |
| 30 | 1.0 | 40.0 | 91.0 | 0.3395 | 29.5 | |
| 85 | 5.833333333333333 | 31.0 | 22.666666666666668 | 0.5155 | 36.833333333333336 | |
| 52 | 2.4545454545454546 | 16.454545454545453 | 91.45454545454545 | 0.48700000000000004 | 25.181818181818183 | 0.27272727 |
| 0 | 3.6285714285714286 | 1.5142857142857142 | 0.0 | 0.38842857142857146 | 30.714285714285715 | 0.457142857 |
| 98 | 5.333333333333333 | 13.666666666666666 | 19.333333333333332 | 0.8506666666666667 | 31.666666666666668 | 0.66666666 |
| 110 | 4.333333333333333 | 33.666666666666664 | 123.33333333333333 | 0.5733333333333334 | 39.0 | 0.66666666 |
| NULL | 1.0 | 23.0 | 94.0 | NULL | NaN | |
| 96 | 4.0 | 0.0 | 0.0 | 0.22975 | 31.5 | |
| 100 | 3.6666666666666665 | 36.666666666666664 | 104.0 | 0.7133333333333333 | 34.0 | 0.33333333 |
| 70 | 4.087719298245614 | 24.05263157894737 | 94.96491228070175 | 0.438578947368421 | 31.982456140350877 | 0.403508771 |
| 61 | 6.0 | 0.0 | 0.0 | 0.151 | 55.0 | |
| 75 | 2.625 | 13.875 | 0.0 | 0.382625 | 33.25 | |
| 46 | 1.5 | 20.0 | 209.0 | 0.41500000000000004 | 22.0 | |
| 78 | 4.888888888888889 | 23.733333333333334 | 66.75555555555556 | 0.4885111111111111 | 39.08888888888889 | 0.377777777 |

only showing top 20 rows